

Markovian Embeddability of $SU(2)$ -Covariant Spin Channels

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Abstract

For a fixed irreducible spin representation, we classify the quantum Markov semigroups that commute with conjugation by $SU(2)$. Their generators form a simplicial cone indexed by the nonzero tensor ranks. We compute the generator eigenvalue matrix in terms of Wigner $6j$ symbols, give its closed inverse, and obtain logarithmic endpoint inequalities for spins 1, $3/2$, and 2. The same cone characterizes the instantaneous logarithmic derivative of the absolute tensor-sector eigenvalues along any invertible differentiable covariant CP-divisible path.

1 Introduction

The embedding problem asks what a single quantum channel reveals about a possible continuous-time evolution: given Φ , does there exist a GKSL generator $\mathcal{L}$ and a time $t > 0$ such that $\Phi = e^{t\mathcal{L}}$? General finite-dimensional criteria must contend with matrix logarithm branches and conditional complete positivity [10]. Symmetry changes that problem substantially when it forces both the channel and the generator to be scalar on a multiplicity-free decomposition.

For irreducible $SU(2)$ representations, the static channel geometry is explicit. Al Nuwairan proved that the EPOSIC channels are precisely the extreme points of the covariant-channel simplex [1, Proposition 4.1]. The same vertices appear as Clebsch–Gordan channels in [4, Sec. 2], where low-rank entanglement and degradability questions are analyzed, in particular in Sec. 4. Cîrstoiu, Korzekwa, and Jennings describe the simplex in Liouville coordinates and derive the action of its extremal channels on irreducible tensor operators [5, Theorem 7 and Appendix B]. Aschieri, Ruba, and Solovej developed quantitative large-spin approximations and trace formulas for this simplex [2, Sec. 1]. Recent work on Weyl dynamical maps gives semigroup and non-Markovianity criteria in a different covariant family [11, Proposition 2]. Rudnicki, Klimov, Muñoz, Leuchs, and Sánchez-Soto have also given a recent Choi–Kraus characterization of the static $SU(2)$ -covariant channel family [9, Abstract].

The continuous-time covariance principle predates the present specialization. Holevo showed that the completely positive part in a standard representation of a norm-continuous covariant semigroup may be chosen covariant and its Hamiltonian may be chosen to commute with the symmetry representation [7, Abstract]; the form-generator theory beyond norm continuity is developed in [8, Abstract]. In the finite-dimensional irreducible spin setting, the multiplicity-free adjoint decomposition resolves Holevo’s covariant completely positive part into one scalar rate per tensor rank. We include that specialization with the normalization needed for the embedding calculation.

Here the symmetry is imposed on the continuous-time generator as well as on its endpoint. Thus “embeddable” below means embeddable by an $SU(2)$ -covariant GKSL generator; no assertion is made about a noncovariant logarithm of the same endpoint. The main results are:

1. every invariant generator is a unique nonnegative combination of the tensor-rank dissipators $\mathcal{D}_k$;
2. its sector eigenvalue matrix $M^{(j)}$ and the inverse matrix are single $6j$ -symbol formulas;

3. positivity of all sector eigenvalues together with $(M^{(j)})^{-1} \log \boldsymbol{\eta} \geq 0$ is necessary and sufficient for endpoint embeddability, yielding explicit facets through spin 2;
4. the same inequalities applied to $d(\log |\boldsymbol{\eta}(t)|)/dt$ characterize all invertible differentiable covariant CP-divisible paths, including paths on fixed negative-spectrum branches.

All low-spin matrices, intersections, and plotted meshes are generated by exact symbolic scripts included with the paper.

2 Tensor conventions and covariant channels

Let $j \in \frac{1}{2}\mathbb{N}_0$, let $\mathcal{H}_j$ be the spin- j irreducible unitary representation, and put $d = 2j + 1$. Conjugation defines the unitary representation

$$\mathcal{U}_g(X) = U_j(g)XU_j(g)^\dagger \quad \text{on } B(\mathcal{H}_j).$$

The Clebsch–Gordan decomposition is multiplicity free:

$$B(\mathcal{H}_j) \cong \mathcal{H}_j \otimes \mathcal{H}_j^* \cong \bigoplus_{\ell=0}^{2j} V_\ell. \quad (1)$$

We use the Hilbert–Schmidt orthonormal spherical tensors of Breuer [3, Appendix A, Eqs. (A1)–(A6)]. In the angular momentum basis they are

$$\langle j, m_1 | T_q^{(k)} | j, m_2 \rangle = \sqrt{2k+1} (-1)^{j-m_1} \begin{pmatrix} j & j & k \\ m_1 & -m_2 & -q \end{pmatrix}. \quad (2)$$

Thus

$$\text{Tr} \left((T_q^{(k)})^\dagger T_{q'}^{(k')} \right) = \delta_{kk'} \delta_{qq'}, \quad T_0^{(0)} = I/\sqrt{d}, \quad (3)$$

$$\mathcal{U}_g(T_q^{(k)}) = \sum_{r=-k}^k D_{rq}^{(k)}(g) T_r^{(k)}, \quad (T_q^{(k)})^\dagger = (-1)^q T_{-q}^{(k)}. \quad (4)$$

Lemma 2.1 (Scalar tensor sum). *For $0 \leq k \leq 2j$,*

$$\sum_{q=-k}^k (T_q^{(k)})^\dagger T_q^{(k)} = \sum_{q=-k}^k T_q^{(k)} (T_q^{(k)})^\dagger = \frac{2k+1}{d} I. \quad (5)$$

Proof. The first sum is invariant under conjugation by (4), hence it is scalar by irreducibility of U_j . Its trace is $2k+1$ by (3), which gives the first identity. The adjoint relation in (4) permutes the summands of the two sums, giving the second identity. $\square$

Definition 2.2 (Invariant dissipators). For $1 \leq k \leq 2j$, define

$$\mathcal{D}_k(X) = \sum_{q=-k}^k \left(T_q^{(k)} X (T_q^{(k)})^\dagger - \frac{1}{2} \{ (T_q^{(k)})^\dagger T_q^{(k)}, X \} \right) \quad (6)$$

$$= \sum_{q=-k}^k T_q^{(k)} X (T_q^{(k)})^\dagger - \frac{2k+1}{d} X. \quad (7)$$

Lemma 2.3 (Dirichlet form). *For every $X \in B(\mathcal{H}_j)$ and $1 \leq k \leq 2j$,*

$$\langle X, \mathcal{D}_k(X) \rangle_{\text{HS}} = -\frac{1}{2} \sum_{q=-k}^k \|[T_q^{(k)}, X]\|_{\text{HS}}^2 \leq 0. \quad (8)$$

Consequently every entry $M_{\ell k}^{(j)}$ introduced below is nonpositive, and every covariantly embeddable channel satisfies $0 < \eta_\ell \leq 1$.

Proof. Write $A_q = T_q^{(k)}$ and $c = (2k+1)/d$. The set $\{A_q\}_{q=-k}^k$ is closed under adjoints up to signs, and Lemma 2.1 gives $\sum_q A_q^\dagger A_q = \sum_q A_q A_q^\dagger = cI$. Expanding the squared commutators therefore yields

$$\frac{1}{2} \sum_q \|[A_q, X]\|_{\text{HS}}^2 = c\|X\|_{\text{HS}}^2 - \text{Re} \sum_q \text{Tr}(X^\dagger A_q X A_q^\dagger).$$

The Kraus sum is Hilbert–Schmidt self-adjoint because $A_q^\dagger = (-1)^q A_{-q}$, so the trace sum on the right is real. Comparison with (7) proves (8). Taking $X = T_m^{(\ell)}$ gives $M_{\ell k}^{(j)} \leq 0$. The final assertion follows from $\eta_\ell = \exp(t \sum_k M_{\ell k}^{(j)} \gamma_k)$. $\square$

Proposition 2.4 (Rank-one angular-momentum diffusion). *Assume $j > 0$, and let J_x, J_y, J_z be the spin- j angular momentum operators. Then*

$$\mathcal{D}_1(X) = -\frac{3}{2j(j+1)(2j+1)} \sum_{a=x,y,z} [J_a, [J_a, X]]. \quad (9)$$

In particular, its eigenvalue on V_ℓ is

$$M_{\ell 1}^{(j)} = -\frac{3\ell(\ell+1)}{2j(j+1)(2j+1)}. \quad (10)$$

Proof. Rotation invariance and $\sum_a J_a^2 = j(j+1)I$ give

$$\text{Tr}(J_a J_b) = s \delta_{ab}, \quad s = \frac{j(j+1)(2j+1)}{3}.$$

With spherical components $J_0 = J_z$ and $J_{\pm 1} = \mp(J_x \pm iJ_y)/\sqrt{2}$, the convention (2) is $T_q^{(1)} = J_q/\sqrt{s}$. Changing unitarily from spherical to Cartesian components in (7) therefore gives

$$\mathcal{D}_1(X) = \frac{1}{s} \left(\sum_a J_a X J_a - j(j+1)X \right),$$

which is (9). The adjoint Casimir $\sum_a [J_a, [J_a, \cdot]]$ acts on the spin- ℓ summand V_ℓ by $\ell(\ell+1)$, proving (10). $\square$

Proposition 2.5 (Depolarizing direction). *Let*

$$\mathcal{P}_0(X) = \frac{\text{Tr}(X)}{d} I. \quad (11)$$

Then

$$\mathcal{P}_0 - \text{id} = \frac{1}{d} \sum_{k=1}^{2j} \mathcal{D}_k. \quad (12)$$

Consequently the equal-rate point $\gamma_k = 1/d$ generates the standard depolarizing semigroup

$$e^{t(\mathcal{P}_0 - \text{id})}(X) = e^{-t}X + (1 - e^{-t}) \frac{\text{Tr}(X)}{d} I, \quad (13)$$

whose nontrivial sector eigenvalues are all e^{-t} .

Proof. The tensors $T_q^{(k)}$, over all k, q , form a Hilbert–Schmidt orthonormal basis of $B(\mathcal{H}_j)$. Completeness of any such matrix basis gives

$$\sum_{k=0}^{2j} \sum_{q=-k}^k T_q^{(k)} X (T_q^{(k)})^\dagger = \text{Tr}(X) I.$$

The $k = 0$ term is X/d , and $\sum_{k=1}^{2j} (2k+1) = d^2 - 1$. Summing (7) over $k \geq 1$ therefore gives $\text{Tr}(X)I - dX$, which is (12). Since $\mathcal{P}_0$ is a projection and $\mathcal{P}_0 \text{id} = \text{id} \mathcal{P}_0 = \mathcal{P}_0$, its exponential is (13). $\square$

For later comparison with the complete channel simplex, set

$$\mathcal{E}_k(X) = \frac{d}{2k+1} \sum_{q=-k}^k T_q^{(k)} X (T_q^{(k)})^\dagger, \quad 0 \leq k \leq 2j. \quad (14)$$

Proposition 2.6 (Covariant-channel simplex). *The maps $\mathcal{E}_0, \dots, \mathcal{E}_{2j}$ are precisely the vertices of the simplex of $\text{SU}(2)$ -covariant channels on $B(\mathcal{H}_j)$. Moreover,*

$$\mathcal{D}_k = \frac{2k+1}{d} (\mathcal{E}_k - \text{id}). \quad (15)$$

Proof. Lemma 2.1 shows that every $\mathcal{E}_k$ is trace preserving and unital. Under column vectorization, its Choi matrix is $d/(2k+1)$ times the orthogonal projector onto the rank- k summand of $\mathcal{H}_j \otimes \mathcal{H}_j^*$. The irreducible Choi projectors are exactly the simplex vertices by [2, Sec. 1, Eqs. (1.34)–(1.37), and Appendix A]. Equation (15) follows by comparing (7) and (14). $\square$

3 Classification of invariant generators

Theorem 3.1 (Invariant-generator decomposition). *A linear map $\mathcal{L} : B(\mathcal{H}_j) \rightarrow B(\mathcal{H}_j)$ generates a norm-continuous $\text{SU}(2)$ -covariant quantum Markov semigroup if and only if there are unique numbers $\gamma_1, \dots, \gamma_{2j} \geq 0$ such that*

$$\mathcal{L} = \sum_{k=1}^{2j} \gamma_k \mathcal{D}_k. \quad (16)$$

In particular, an invariant generator has no nonzero Hamiltonian part and is self-adjoint for the Hilbert–Schmidt inner product.

The covariance statement behind this decomposition is the finite-dimensional compact-group case of Holevo’s standard-form result [7, Abstract]. The proof below identifies the unique tensor-rank coefficients in the convention (3).

Proof. Choose the tensors $T_q^{(k)}$, $1 \leq k \leq 2j$, as an orthonormal basis of the traceless operators. The canonical finite-dimensional Gorini–Kossakowski–Sudarshan representation [6, Theorem 2.2 and Lemma 2.3] gives unique data $H = H^\dagger$, $\text{Tr} H = 0$, and a positive semidefinite matrix C on the traceless operator space such that

$$\mathcal{L}(X) = -i[H, X] + \sum_{a,b} C_{ab} \left(F_a X F_b^\dagger - \frac{1}{2} \{F_b^\dagger F_a, X\} \right), \quad (17)$$

where $\{F_a\}$ is the chosen tensor basis.

For $g \in \text{SU}(2)$, conjugating (17) by $\mathcal{U}_g$ replaces H by $U_j(g) H U_j(g)^\dagger$ and C by its conjugate under the representation of $\text{SU}(2)$ on the traceless operator space. Covariance of $\mathcal{L}$, followed by uniqueness of the canonical data, therefore gives

$$U_j(g) H U_j(g)^\dagger = H, \quad C \mathcal{U}_g = \mathcal{U}_g C \quad (g \in \text{SU}(2)). \quad (18)$$

Schur's lemma and irreducibility of U_j make H scalar; the condition $\text{Tr } H = 0$ gives $H = 0$.

By (1), the traceless operator space is the multiplicity-free sum $\bigoplus_{k=1}^{2j} V_k$. A second application of Schur's lemma gives

$$C = \bigoplus_{k=1}^{2j} \gamma_k I_{V_k}. \quad (19)$$

Positivity of C is equivalent to $\gamma_k \geq 0$ for every k . Substitution of (19) into (17) gives (16).

Conversely, each $\mathcal{D}_k$ is in GKSL form and is covariant by (4). Hence every nonnegative combination in (16) generates a covariant quantum Markov semigroup. Uniqueness of the canonical GKS matrix gives uniqueness of the coefficients. Finally, the adjoint relation in (4) shows that the Kraus sum in (7) is Hilbert–Schmidt self-adjoint. $\square$

4 The exact $6j$ -symbol matrix

Lemma 4.1 (Sector diagonalization and reality). *If $A : B(\mathcal{H}_j) \rightarrow B(\mathcal{H}_j)$ is covariant, then there are scalars $a_0, \dots, a_{2j}$ such that*

$$A(T_m^{(\ell)}) = a_\ell T_m^{(\ell)} \quad (-\ell \leq m \leq \ell). \quad (20)$$

If A is Hermiticity preserving, every a_ℓ is real. If A is a channel, then $a_0 = 1$.

Proof. The irreducible summands in (1) are pairwise inequivalent and occur with multiplicity one, so (20) follows from Schur's lemma. Since $(T_0^{(\ell)})^\dagger = T_0^{(\ell)}$, Hermiticity preservation forces $a_\ell T_0^{(\ell)} = \overline{a_\ell} T_0^{(\ell)}$, hence $a_\ell \in \mathbb{R}$. Finally, covariance makes $A(I)$ invariant under U_j , so $A(I) = cI$; trace preservation gives $c = 1$. $\square$

For a covariant generator, write $\mathcal{L}(T_m^{(\ell)}) = \lambda_\ell T_m^{(\ell)}$.

Lemma 4.2 (Tensor recoupling). *For $0 \leq k, \ell \leq 2j$,*

$$\sum_{q=-k}^k T_q^{(k)} T_m^{(\ell)} (T_q^{(k)})^\dagger = a_{\ell k} T_m^{(\ell)}, \quad (21)$$

where

$$a_{\ell k} = (2k+1)(-1)^{2j+k+\ell} \left\{ \begin{matrix} j & j & k \\ j & j & \ell \end{matrix} \right\}. \quad (22)$$

Proof. The left side of (21) defines a covariant map, so it is scalar on V_ℓ . We calculate the scalar with the invariant unitary

$$\beta : \mathcal{H}_j^* \longrightarrow \mathcal{H}_j, \quad \beta |j, m\rangle = (-1)^{j-m} |j, -m\rangle. \quad (23)$$

For column vectorization into $\mathcal{H}_j \otimes \mathcal{H}_j^*$, the Kraus map in (21) is represented by $\sum_q T_q^{(k)} \otimes \overline{T_q^{(k)}}$. The tensors are real, and Breuer's Eq. (A6) states $\beta(T_q^{(k)})^\top \beta^\dagger = (-1)^k T_q^{(k)}$. Taking adjoints gives

$$\beta \overline{T_q^{(k)}} \beta^\dagger = (-1)^k (T_q^{(k)})^\dagger. \quad (24)$$

Consequently, after applying $I \otimes \beta$, the transfer matrix of the Kraus map is $(-1)^k Q_k$, where

$$Q_k = \sum_{q=-k}^k T_q^{(k)} \otimes (T_q^{(k)})^\dagger$$

is exactly Breuer's invariant tensor [3, Eq. (2.7)]. Let P_ℓ be the projector onto the coupled spin- ℓ summand. It follows that

$$a_{\ell k} = (-1)^k \frac{\text{Tr}(Q_k P_\ell)}{2\ell + 1}. \quad (25)$$

Breuer's Eqs. (2.13)–(2.14), with $K = k$, $J = \ell$, and $j_1 = j_2 = j$, state

$$\text{Tr}(Q_k P_\ell) = \sqrt{(2k+1)(2\ell+1)} L_{k\ell}, \quad (26)$$

$$L_{k\ell} = \sqrt{(2k+1)(2\ell+1)} (-1)^{2j+\ell} \begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix}. \quad (27)$$

Substitution in (25), followed by the tetrahedral symmetry of the $6j$ symbol, gives (22). $\square$

Corollary 4.3 (Channel-vertex eigenvalues). *The simplex vertex $\mathcal{E}_k$ has tensor-sector eigenvalues*

$$v_{\ell k}^{(j)} = (2j+1)(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}, \quad 0 \leq k, \ell \leq 2j. \quad (28)$$

In particular, $v_{0k}^{(j)} = v_{\ell 0}^{(j)} = 1$.

Proof. Multiply (22) by the normalization $d/(2k+1)$ in (14). The cases $k = 0$ and $\ell = 0$ also follow directly from $\mathcal{E}_0 = \text{id}$ and trace preservation. $\square$

Theorem 4.4 (Generator eigenvalue matrix and inverse). *For $1 \leq \ell, k \leq 2j$, set*

$$M_{\ell k}^{(j)} = (2k+1) \left[(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix} - \frac{1}{2j+1} \right]. \quad (29)$$

If $\mathcal{L} = \sum_k \gamma_k \mathcal{D}_k$, then

$$(\lambda_1, \dots, \lambda_{2j})^\top = M^{(j)} (\gamma_1, \dots, \gamma_{2j})^\top. \quad (30)$$

The matrix is invertible, with

$$((M^{(j)})^{-1})_{k\ell} = (2\ell+1)(-1)^{2j+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}. \quad (31)$$

Moreover,

$$|\det M^{(j)}| = 2j+1. \quad (32)$$

Proof. Equations (7) and (22) give (29) and (30). It remains to invert the matrix.

Put $n = 2j$, $w_r = 2r+1$, and let $V = (v_{\ell k}^{(j)})_{0 \leq \ell, k \leq n}$ be the full vertex eigenvalue matrix from Corollary 4.3. Let $W = \text{diag}(w_0, \dots, w_n)$ and $S = \text{diag}((-1)^0, \dots, (-1)^n)$. Breuer's recoupling matrix is

$$L_{k\ell} = \sqrt{w_k w_\ell} (-1)^{n+\ell} \begin{Bmatrix} j & j & \ell \\ j & j & k \end{Bmatrix};$$

it is orthogonal by the sum rule following [3, Eq. (2.14)]. Tetrahedral symmetry and (28) give

$$V = (n+1)W^{-1/2}L^\top W^{-1/2}S, \quad V^{-1} = \frac{1}{n+1}SW^{1/2}LW^{1/2}. \quad (33)$$

Hence

$$(V^{-1})_{k\ell} = \frac{w_k w_\ell}{n+1} (-1)^{n+k+\ell} \begin{Bmatrix} j & j & k \\ j & j & \ell \end{Bmatrix}. \quad (34)$$

Since the first row and first column of V consist of ones, subtracting its first row from all other rows gives the block matrix

$$\begin{pmatrix} 1 & \mathbf{1}^\top \\ 0 & B \end{pmatrix}, \quad B_{\ell k} = v_{\ell k}^{(j)} - 1 \quad (1 \leq \ell, k \leq n).$$

The lower-right block of V^{-1} is therefore B^{-1} , while (15) says $M^{(j)} = B \operatorname{diag}(w_1/d, \dots, w_n/d)$. Combining this identity with (34) proves (31).

Finally, (33) gives

$$|\det V| = \frac{d^d}{\prod_{r=0}^n w_r}.$$

Row subtraction shows $\det V = \det B$, and multiplying the columns of B by w_k/d , $1 \leq k \leq n$, gives $|\det M^{(j)}| = d$, proving (32). $\square$

Corollary 4.5 (Dissipator–vertex conversion). *The vertex and generator eigenmatrices satisfy*

$$M_{\ell k}^{(j)} = \frac{2k+1}{2j+1} (v_{\ell k}^{(j)} - 1). \quad (35)$$

Proof. Restrict (15) to V_ℓ . $\square$

Corollary 4.6 (Complete-positivity halfspaces). *Let A be a covariant, Hermiticity-preserving, trace-preserving map with sector eigenvalues $\eta_0 = 1, \eta_1, \dots, \eta_{2j}$. Define*

$$p_k = \frac{2k+1}{2j+1} \sum_{\ell=0}^{2j} (2\ell+1) (-1)^{2j+k+\ell} \left\{ \begin{matrix} j & j & k \\ j & j & \ell \end{matrix} \right\} \eta_\ell, \quad 0 \leq k \leq 2j. \quad (36)$$

Then A is completely positive if and only if $p_k \geq 0$ for every k . In that case $\sum_k p_k = 1$ and

$$A = \sum_{k=0}^{2j} p_k \mathcal{E}_k. \quad (37)$$

Proof. In sector coordinates, $\boldsymbol{\eta} = V\boldsymbol{p}$, where V is the full vertex matrix used in (33). Formula (34) says that $\boldsymbol{p} = V^{-1}\boldsymbol{\eta}$ is exactly (36). Proposition 2.6 identifies complete positivity with nonnegative barycentric weights. The equality $\sum_k p_k = \eta_0 = 1$ follows from the first row of V . $\square$

5 Endpoint embeddability

Theorem 5.1 (Covariant Markovian embedding criterion). *Let Φ be an $\mathrm{SU}(2)$ -covariant channel on $B(\mathcal{H}_j)$, with tensor-sector eigenvalues $\eta_0 = 1, \eta_1, \dots, \eta_{2j}$. There are $t > 0$ and an $\mathrm{SU}(2)$ -covariant GKSL generator $\mathcal{L}$ such that $\Phi = e^{t\mathcal{L}}$ if and only if*

$$\eta_\ell > 0 \quad (1 \leq \ell \leq 2j), \quad (M^{(j)})^{-1} \begin{pmatrix} \log \eta_1 \\ \vdots \\ \log \eta_{2j} \end{pmatrix} \in \mathbb{R}_{\geq 0}^{2j}. \quad (38)$$

Proof. If $\Phi = e^{t\mathcal{L}}$, Theorem 3.1 gives $\mathcal{L} = \sum_k \gamma_k \mathcal{D}_k$ with $\gamma_k \geq 0$. Its sector eigenvalues are real by Theorem 4.4, hence $\eta_\ell = e^{t\lambda_\ell} > 0$, and

$$\log \boldsymbol{\eta} = M^{(j)}(t\boldsymbol{\gamma}).$$

This proves necessity. Conversely, define $r = (M^{(j)})^{-1} \log \boldsymbol{\eta} \geq 0$ and $\mathcal{L} = \sum_k r_k \mathcal{D}_k$. Theorem 3.1 makes $\mathcal{L}$ a covariant GKSL generator. The maps $e^\mathcal{L}$ and Φ have the same scalar on every summand in (1); therefore they are equal. $\square$

Corollary 5.2 (General facets and multiplicative geometry). *For a covariant channel with positive sector eigenvalues, define*

$$r_k = \sum_{\ell=1}^{2j} (2\ell+1)(-1)^{2j+k+\ell} \left\{ \begin{matrix} j & j & k \\ j & j & \ell \end{matrix} \right\} \log \eta_\ell, \quad 1 \leq k \leq 2j. \quad (39)$$

Then the channel is covariantly embeddable if and only if the $2j$ inequalities $r_k \geq 0$ hold. The vector r is the unique integrated-rate vector, and the facet $r_k = 0$ is the image of the generator face on which the rank- k dissipator is absent.

The embeddable eigenvalue set is closed under componentwise multiplication and under weighted geometric interpolation: if $\boldsymbol{\eta}$ and $\boldsymbol{\zeta}$ are embeddable and $0 \leq s \leq 1$, then so are

$$\boldsymbol{\eta}\boldsymbol{\zeta} \quad \text{and} \quad \boldsymbol{\eta}^s \boldsymbol{\zeta}^{1-s}, \quad (40)$$

where all operations are componentwise.

Proof. Equation (39) is $(M^{(j)})^{-1} \log \boldsymbol{\eta}$ written using (31), so the first assertion is Theorem 5.1. In logarithmic coordinates, the two operations in (40) add the integrated rates and take their convex combinations, respectively. Both preserve $\mathbb{R}_{\geq 0}^{2j}$. Equivalently, all invariant generators commute because they are diagonal on the tensor-rank sectors, so the corresponding exponentials compose by addition of their rates. $\square$

Proposition 5.3 (Poisson collision representation). *For $r_k \geq 0$, put $a_k = (2k+1)r_k/d$. The semigroup endpoint with integrated rates $r = (r_1, \dots, r_{2j})$ factors as*

$$\exp \left(\sum_{k=1}^{2j} r_k \mathcal{D}_k \right) = \prod_{k=1}^{2j} \left[e^{-a_k} \sum_{m=0}^{\infty} \frac{a_k^m}{m!} \mathcal{E}_k^m \right]. \quad (41)$$

Thus each extreme generator ray is the continuous-time Poissonization of repeated applications of one Clebsch–Gordan vertex channel, and a general invariant semigroup is the product of these commuting processes.

Proof. Sector diagonalization makes the maps $\mathcal{D}_k$ commute. Equation (15) gives

$$e^{r_k \mathcal{D}_k} = e^{a_k (\mathcal{E}_k - \text{id})} = e^{-a_k} \sum_{m=0}^{\infty} \frac{a_k^m}{m!} \mathcal{E}_k^m.$$

The series converges in operator norm. Multiplying these identities and using commutativity proves (41). $\square$

Corollary 5.4 (Spin 1/2 baseline). *For $j = 1/2$, the covariant-channel interval is $-1/3 \leq \eta_1 \leq 1$, while*

$$M^{(1/2)} = (-2), \quad r_1 = -\frac{1}{2} \log \eta_1. \quad (42)$$

Consequently a spin-1/2 covariant channel is covariantly embeddable if and only if $0 < \eta_1 \leq 1$. Thus every positive-spectrum channel in this one-dimensional simplex is embeddable.

Proof. The two channel vertices have eigenvalues 1 and $-1/3$, by (28). Equation (29) gives $M^{(1/2)} = -2$, and Theorem 5.1 gives (42). $\square$

Corollary 5.5 (Spin 1). *For $j = 1$,*

$$M^{(1)} = \begin{pmatrix} -\frac{1}{2} & -\frac{5}{2} \\ -\frac{3}{2} & -\frac{3}{2} \end{pmatrix}, \quad (M^{(1)})^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{5}{6} \\ -\frac{1}{2} & \frac{1}{6} \end{pmatrix}. \quad (43)$$

Writing $x_\ell = \log \eta_\ell$, a covariant channel is covariantly embeddable if and only if $\eta_1, \eta_2 > 0$ and

$$3x_1 - 5x_2 \geq 0, \quad -3x_1 + x_2 \geq 0. \quad (44)$$

Equivalently,

$$\eta_1^3 \geq \eta_2^5, \quad \eta_2 \geq \eta_1^3. \quad (45)$$

Proof. Substitution of the four relevant $6j$ symbols into (29) gives (43). The two rows of its inverse, after multiplication by the positive denominator 6, give (44). Exponentiation gives (45). $\square$

Corollary 5.6 (Spin $3/2$). For $j = 3/2$,

$$M^{(3/2)} = \begin{pmatrix} -\frac{1}{23} & -1 & -\frac{14}{5} \\ -\frac{1}{23} & -2 & -\frac{7}{5} \\ -\frac{5}{8} & -1 & -\frac{9}{5} \end{pmatrix} \quad (46)$$

and

$$(M^{(3/2)})^{-1} = \frac{1}{20} \begin{pmatrix} 11 & 5 & -21 \\ 3 & -15 & 7 \\ -9 & 5 & -1 \end{pmatrix}. \quad (47)$$

Writing $x_\ell = \log \eta_\ell$, a covariant channel is covariantly embeddable if and only if every $\eta_\ell > 0$ and

$$11x_1 + 5x_2 - 21x_3 \geq 0, \quad (48)$$

$$3x_1 - 15x_2 + 7x_3 \geq 0, \quad (49)$$

$$-9x_1 + 5x_2 - x_3 \geq 0. \quad (50)$$

Equivalently,

$$\eta_1^{11} \eta_2^5 \geq \eta_3^{21}, \quad \eta_1^3 \eta_3^7 \geq \eta_2^{15}, \quad \eta_2^5 \geq \eta_1^9 \eta_3. \quad (51)$$

Proof. Substitution in (29) gives (46); exact matrix inversion gives (47). The rows of the inverse give the three inequalities, and exponentiation gives (51). $\square$

Corollary 5.7 (Spin 2). For $j = 2$,

$$M^{(2)} = \begin{pmatrix} -\frac{1}{10} & -\frac{1}{7} & -\frac{7}{11} & -\frac{3}{9} \\ -\frac{1}{10} & -\frac{1}{14} & -\frac{1}{5} & -\frac{9}{7} \\ -\frac{3}{5} & -\frac{1}{7} & -\frac{9}{10} & -\frac{27}{14} \\ -1 & -\frac{5}{7} & -\frac{3}{2} & -\frac{25}{14} \end{pmatrix} \quad (52)$$

and

$$(M^{(2)})^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & -\frac{6}{5} \\ \frac{3}{10} & -\frac{3}{14} & -\frac{4}{5} & \frac{18}{35} \\ 0 & -\frac{4}{7} & \frac{1}{2} & -\frac{9}{70} \\ -\frac{2}{5} & \frac{2}{7} & -\frac{1}{10} & \frac{1}{70} \end{pmatrix}. \quad (53)$$

Writing $x_\ell = \log \eta_\ell$, a covariant channel is covariantly embeddable if and only if every $\eta_\ell > 0$ and

$$5x_1 + 5x_2 - 12x_4 \geq 0, \quad (54)$$

$$21x_1 - 15x_2 - 56x_3 + 36x_4 \geq 0, \quad (55)$$

$$-40x_2 + 35x_3 - 9x_4 \geq 0, \quad (56)$$

$$-28x_1 + 20x_2 - 7x_3 + x_4 \geq 0. \quad (57)$$

Equivalently,

$$\eta_1^5 \eta_2^5 \geq \eta_4^{12}, \quad \eta_1^{21} \eta_4^{36} \geq \eta_2^{15} \eta_3^{56}, \quad (58)$$

$$\eta_3^{35} \geq \eta_2^{40} \eta_4^9, \quad \eta_2^{20} \eta_4 \geq \eta_1^{28} \eta_3^7. \quad (59)$$

Proof. Equations (29) and (31) give (52) and (53). Multiplying the four inverse rows by 10, 70, 70, 70, respectively, and dividing by their positive common factors gives the primitive facet normals in (54)–(57). Exponentiation gives (58)–(59). $\square$

6 Low-spin channel geometry

Proposition 6.1 (Low-spin complete-positivity facets). *For spin 1, a real pair (η_1, η_2) is the sector spectrum of a covariant channel if and only if*

$$1 + 3\eta_1 + 5\eta_2 \geq 0, \quad (60)$$

$$2 + 3\eta_1 - 5\eta_2 \geq 0, \quad (61)$$

$$2 - 3\eta_1 + \eta_2 \geq 0. \quad (62)$$

For spin 3/2, the corresponding four inequalities are

$$1 + 3\eta_1 + 5\eta_2 + 7\eta_3 \geq 0, \quad (63)$$

$$5 + 11\eta_1 + 5\eta_2 - 21\eta_3 \geq 0, \quad (64)$$

$$5 + 3\eta_1 - 15\eta_2 + 7\eta_3 \geq 0, \quad (65)$$

$$5 - 9\eta_1 + 5\eta_2 - \eta_3 \geq 0. \quad (66)$$

Proof. For spin 1, the three barycentric coordinates (36) are

$$\frac{1 + 3\eta_1 + 5\eta_2}{9}, \quad \frac{2 + 3\eta_1 - 5\eta_2}{6}, \quad \frac{5(2 - 3\eta_1 + \eta_2)}{18}.$$

For spin 3/2, they are

$$\begin{aligned} & \frac{1 + 3\eta_1 + 5\eta_2 + 7\eta_3}{16}, \quad \frac{3(5 + 11\eta_1 + 5\eta_2 - 21\eta_3)}{80}, \\ & \frac{5 + 3\eta_1 - 15\eta_2 + 7\eta_3}{16}, \quad \frac{7(5 - 9\eta_1 + 5\eta_2 - \eta_3)}{80}. \end{aligned}$$

Corollary 4.6 proves the claim. $\square$

Proposition 6.2 (Exact comparison polytopes). *In tensor-eigenvalue coordinates, the spin-1 covariant-channel simplex and its positive-spectrum part are*

$$S_1 = \text{conv} \left\{ (1, 1), \left(\frac{1}{2}, -\frac{1}{2} \right), \left(-\frac{1}{2}, \frac{1}{10} \right) \right\}, \quad (67)$$

$$S_1^+ = \text{conv} \left\{ (0, 0), \left(0, \frac{2}{5} \right), \left(\frac{2}{3}, 0 \right), (1, 1) \right\}. \quad (68)$$

For spin 3/2, the full simplex is

$$\begin{aligned} S_{3/2} = \text{conv} \{ & (1, 1, 1), \left(\frac{11}{15}, \frac{1}{5}, -\frac{3}{5} \right), \left(\frac{1}{5}, -\frac{3}{5}, \frac{1}{5} \right), \\ & \left(-\frac{3}{5}, \frac{1}{5}, -\frac{1}{35} \right) \}, \end{aligned} \quad (69)$$

and its positive-spectrum part has the eight vertices

$$\begin{aligned} & (0, 0, 0), \quad \left(0, 0, \frac{5}{21} \right), \quad \left(0, \frac{1}{3}, 0 \right), \quad \left(0, \frac{1}{2}, \frac{5}{14} \right), \\ & \left(\frac{1}{2}, 0, \frac{1}{2} \right), \quad \left(\frac{5}{9}, 0, 0 \right), \quad \left(\frac{5}{6}, \frac{1}{2}, 0 \right), \quad (1, 1, 1). \end{aligned} \quad (70)$$

Proof. Substitution of $j = 1$ and $j = 3/2$ in (28) gives (67) and (69). For the positive-spectrum intersections, write a point as $\eta = V\lambda$, where the columns of V are the displayed simplex vertices and $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$. A vertex of the intersection with $\eta_\ell \geq 0$ is the unique solution of a full-rank selection of the bounding equalities $\lambda_k = 0$ and $\eta_\ell = 0$. The exact enumeration in `code/generate_plot_data.py` solves every such rational system and tests all remaining inequalities. Its certificate `paper/generated/geometry_exact.txt` is (68) and (70); exhaustion of all full-rank active sets proves that there are no further vertices. $\square$

Proposition 6.3 (Euclidean volume of the embeddable region). *Let $n = 2j$, let $\mathbf{M}_j \subset \mathbb{R}^n$ be the set of covariantly embeddable sector spectra, and define*

$$c_k = - \sum_{\ell=1}^n M_{\ell k}^{(j)} > 0. \quad (71)$$

With respect to Lebesgue measure in the fixed sector-eigenvalue coordinates,

$$\text{vol}_n(\mathbf{M}_j) = \frac{2j+1}{\prod_{k=1}^n c_k}. \quad (72)$$

For spins 1, 3/2, and 2, the column-decay vectors and volumes are

j	$(c_1, \dots, c_{2j})$	$\text{vol}_{2j}(\mathbf{M}_j)$
1	(2, 4)	3/8
3/2	(2, 4, 6)	1/12
2	(2, 4, 6, 8)	5/384.

(73)

Moreover,

$$\text{area}(\mathbf{S}_1) = \frac{9}{10}, \quad \text{area}(\mathbf{S}_1^+) = \frac{8}{15}, \quad (74)$$

$$\text{vol}_3(\mathbf{S}_{3/2}) = \frac{128}{315}, \quad \text{vol}_3(\mathbf{S}_{3/2}^+) = \frac{25}{126}. \quad (75)$$

Thus the embeddable fractions of the positive-spectrum regions are 45/64 for spin 1 and 21/50 for spin 3/2; the fractions of the full channel simplices are 5/12 and 105/512, respectively.

Proof. By Theorem 5.1, the map

$$F : \mathbb{R}_{\geq 0}^n \longrightarrow \mathbf{M}_j, \quad F(r) = \exp(M^{(j)} r)$$

is one-to-one and onto, with componentwise exponential. In the interior its Jacobian is

$$DF(r) = \text{diag}(F(r)) M^{(j)}.$$

Lemma 2.3 makes every entry of $M^{(j)}$ nonpositive, and invertibility rules out a zero column, so every c_k in (71) is positive. Equations (32) and (71) give

$$|\det DF(r)| = (2j+1) \exp \left(- \sum_{k=1}^n c_k r_k \right).$$

Integrating this product over $\mathbb{R}_{\geq 0}^n$ proves (72).

Summing the columns of the low-spin matrices gives (73). The first entries in (74) and (75) are simplex determinants. The spin-1 positive area follows from the shoelace formula applied to (68). Triangulating each exact face in (70) from an interior point gives 25/126; the rational active-set script records the same determinant sum. Division gives the four stated fractions. $\square$

Figures 1 and 2 compare these polytopes with the exponential image of the generator cone; the zero-spectrum limit is shown only as part of its closure. The spin-1 boundary curves are exact. For spin 3/2, the rendered boundary meshes truncate the nonnegative rates at 8; the inequalities (48)–(50) remain the exact description.

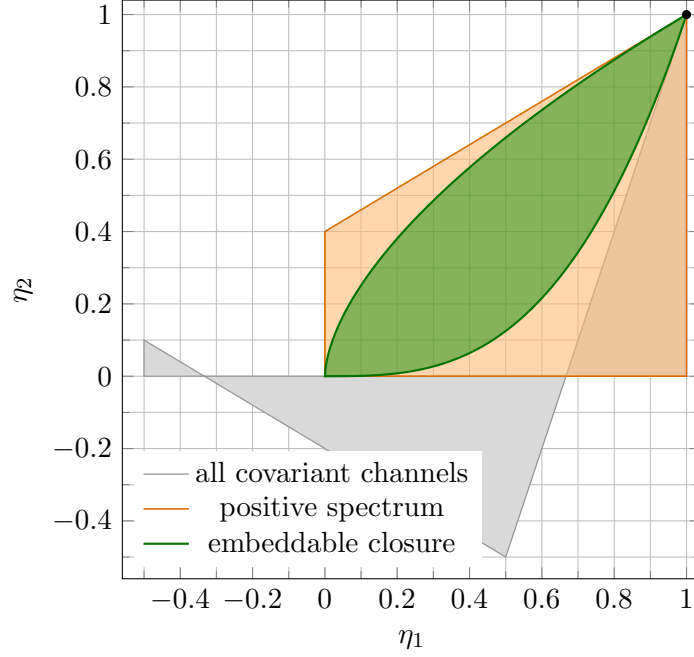


Figure 1: Spin 1. The gray triangle is the exact covariant-channel simplex. The orange polygon is its intersection with $\eta_1, \eta_2 \geq 0$. The green region is the closure of the covariantly embeddable set and is bounded by $\eta_2 = \eta_1^3$ and $\eta_2 = \eta_1^{3/5}$, the two extreme rays of the logarithmic generator cone; $(0, 0)$ itself is not a finite-time endpoint.

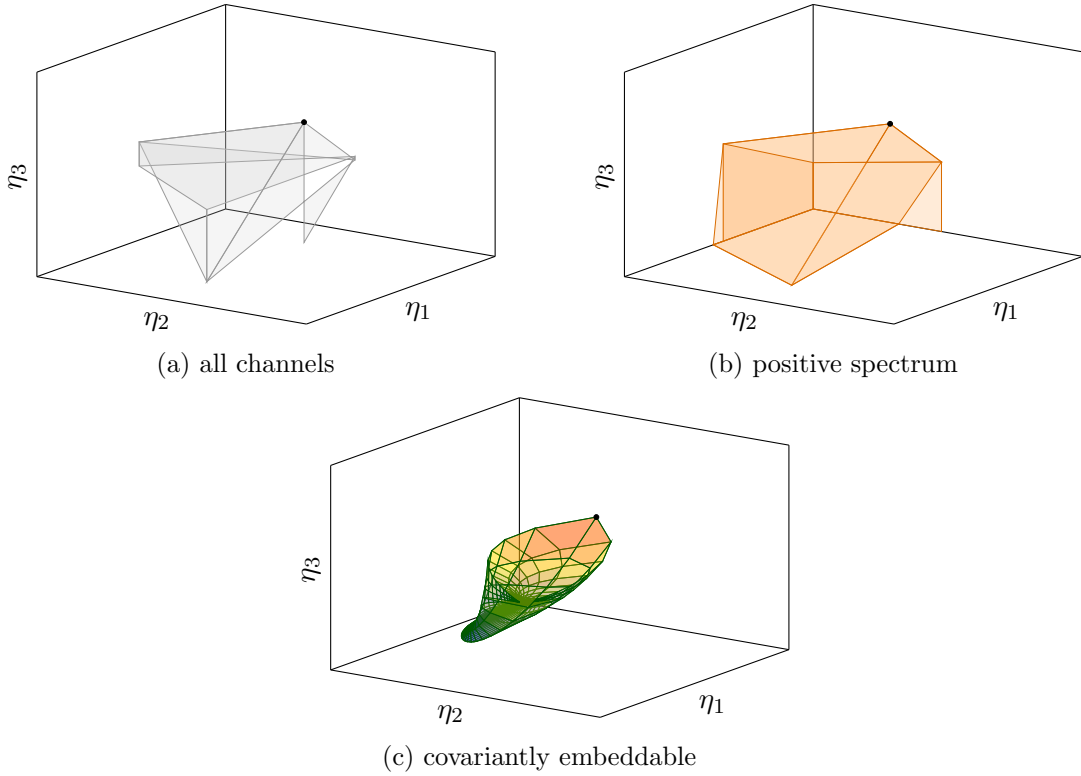


Figure 2: Spin 3/2. Panels (a) and (b) are generated from exact rational polytope vertices. Panel (c) renders the three boundary surfaces obtained by setting one canonical rate to zero; each displayed surface uses rates in $[0, 8]$, while the defining cone is unbounded.

7 Differentiable CP-divisible paths

Definition 7.1 (CP-divisibility). A family of channels $\{\Phi_t\}_{t \in I}$ is CP-divisible if for every $s \leq t$ in I there is a channel $V_{t,s}$ such that $\Phi_t = V_{t,s}\Phi_s$.

Lemma 7.2 (Finite-dimensional local criterion). *Let $t \mapsto \Phi_t$ be an invertible C^1 path of channels on a finite-dimensional matrix algebra. It is CP-divisible if and only if*

$$K_t = \dot{\Phi}_t \Phi_t^{-1} \quad (76)$$

is a GKSL generator for every t .

Proof. If the path is CP-divisible, invertibility makes the propagator unique: $V_{t+h,t} = \Phi_{t+h}\Phi_t^{-1}$. It is a channel and

$$V_{t+h,t} = \text{id} + hK_t + o(h).$$

The tangent-cone form of the finite-dimensional GKS theorem [6, Theorem 2.2 and its conditional-complete-positivity proof] says exactly that K_t is a GKSL generator.

Conversely, suppose every K_t is a GKSL generator. Put $\Delta = (t-s)/n$. The chronological Euler products

$$V_{t,s}^{(n)} = \exp(\Delta K_{s+(n-1)\Delta}) \cdots \exp(\Delta K_{s+\Delta}) \exp(\Delta K_s)$$

converge in operator norm to the solution of $\partial_t V_{t,s} = K_t V_{t,s}$, $V_{s,s} = \text{id}$. Indeed, if $B = \sup_{u \in [s,t]} \|K_u\|$ and

$$\omega_K(\delta) = \sup_{\substack{u,v \in [s,t] \\ |u-v| \leq \delta}} \|K_u - K_v\|,$$

then variation of constants between the exact solution and the piecewise-constant evolution gives

$$\|V_{t,s} - V_{t,s}^{(n)}\| \leq (t-s)e^{B(t-s)}\omega_K(\Delta) \longrightarrow 0.$$

Every factor is a channel, and the cone of completely positive maps and the trace-preserving affine subspace are operator-norm closed. Hence $V_{t,s}$ is a channel. Finally, both Φ_t and $V_{t,s}\Phi_s$ solve $\dot{Y}_t = K_t Y_t$ with initial value $Y_s = \Phi_s$; uniqueness for finite-dimensional linear ODEs gives $\Phi_t = V_{t,s}\Phi_s$. $\square$

Theorem 7.3 (Instantaneous logarithmic cone). *Let $t \mapsto \Phi_t$ be an invertible C^1 path of $\text{SU}(2)$ -covariant channels, with tensor-sector eigenvalues $\eta_\ell(t)$. Then the path is CP-divisible if and only if*

$$(M^{(j)})^{-1} \frac{d}{dt} \begin{pmatrix} \log |\eta_1(t)| \\ \vdots \\ \log |\eta_{2j}(t)| \end{pmatrix} \in \mathbb{R}_{\geq 0}^{2j} \quad \text{for every } t. \quad (77)$$

If $\Phi_{t_0} = \text{id}$ at some t_0 , then every $\eta_\ell(t) > 0$ and the absolute values may be omitted.

Proof. Lemma 4.1 makes every $\eta_\ell(t)$ real, while invertibility makes it nonzero. Since the parameter set is an interval, continuity makes the sign of each η_ℓ constant. At an identity point all these signs are positive. Both $\dot{\Phi}_t$ and Φ_t^{-1} commute with every $\mathcal{U}_g$, so the time-local generator K_t in (76) is covariant. On V_ℓ it acts by

$$\frac{\dot{\eta}_\ell(t)}{\eta_\ell(t)} = \frac{d}{dt} \log |\eta_\ell(t)|.$$

Apply Lemma 7.2, then Theorems 3.1 and 4.4. $\square$

Corollary 7.4 (Spin-1/2 negative-spectrum branches). *Let $t \mapsto \Phi_t$ be an invertible C^1 path of spin-1/2 covariant channels. Then it is CP-divisible if and only if*

$$\frac{d}{dt} \log |\eta_1(t)| \leq 0.$$

In particular, the constant path at the nonidentity vertex $\eta_1 = -1/3$ is invertible and CP-divisible.

Proof. Corollary 5.4 gives $(M^{(1/2)})^{-1} = -1/2$, so Theorem 7.3 reduces to the displayed inequality. The constant vertex path has zero logarithmic derivative, and its propagators are the identity channel. $\square$

8 Exact computation and reproducibility

The symbolic calculation has two independent inputs: the Wigner $6j$ formula (29), and the tensor matrices in (2). This separation makes the phase and normalization checks observable rather than folding them into a single table lookup.

Proposition 8.1 (Exact reconstruction protocol). *For $2j = 1, 2, 3, 4$, the repository’s exact-cone script performs all of the following operations in symbolic arithmetic:*

1. *constructs every $T_q^{(k)}$ once from Clebsch–Gordan coefficients and once from the $3j$ formula (2), and tests entrywise equality;*
2. *constructs the matrix of β in (23), tests its unitarity, and tests (24) for every k, q ;*
3. *applies every Kraus sum directly to $T_0^{(\ell)}$, tests that its residual after projection onto $T_0^{(\ell)}$ vanishes, and tests that the extracted scalar equals (29);*
4. *tests (35), the closed inverse (31), and $|\det M^{(j)}| = 2j + 1$.*

The primitive integer rows printed by the script are exactly the facet normals in the spin-1, spin-3/2, and spin-2 corollaries.

Proof. The tensor and Wigner routines return SymPy algebraic numbers, and the tests use exact matrix equality after symbolic simplification. The direct Kraus reconstruction does not call the $6j$ -based generator routine. For each (ℓ, k) , it first checks the matrix identity $\mathcal{D}_k(T_0^{(\ell)}) = m_{\ell k} T_0^{(\ell)}$, including the full residual, and then compares $m_{\ell k}$ with (29). Thus items 1–3 compare the two derivations entry by entry; item 4 substitutes the resulting matrices into the displayed algebraic identities. The executable check is

```
python3 -u code/exact_cones.py 1 2 3 4 --verify.
```

$\square$

Proposition 8.2 (Exact polytope certificates). *The script `code/generate_plot_data.py` enumerates every full-rank active set of the spin-1 and spin-3/2 simplex halfspaces together with $\eta_\ell \geq 0$, solves the systems over the rationals, and records each feasible vertex, its barycentric coordinates, the exact polytope volumes, and the embeddable-region Jacobian volumes in `paper/generated/geometry_exact.txt`. The only floating-point data produced by the script are plotting coordinates: the spin-1 curves and the truncated spin-3/2 boundary meshes.*

Proof. There are finitely many choices of as many active inequalities as the ambient dimension. A polytope vertex occurs in at least one full-rank choice. The script iterates over all choices, discards singular systems, solves the rest over $\mathbb{Q}$, and retains exactly the points that satisfy every halfspace. It then computes barycentric coordinates by the inverse augmented vertex matrix.

Boundary cycles are ordered by exact signs of rational planar cross products. The certificate records those cycles and every rational determinant term in the shoelace and pyramid sums, so no floating-point decision enters a stated area or volume. This is the active-set argument used in Proposition 6.2. $\square$

From the repository root, `make check` runs the exact tensor audit, `make figures` regenerates all plot data and certificates, and `make paper` rebuilds the figures, bibliography, and paper.

9 Conclusion

For a fixed irreducible spin, covariance converts the canonical GKS matrix into one nonnegative rate per nonzero tensor rank. Recoupling then converts those rates to sector decay exponents by the matrix $M^{(j)}$, while orthogonality of the Racah transform supplies the closed inverse (31). Endpoint embedding is therefore a finite set of linear inequalities in logarithmic eigenvalues, and time-dependent CP-divisibility is the pointwise version of the same test applied to logarithms of absolute sector eigenvalues. The low-spin corollaries and comparison plots make the distinction between complete positivity, positive spectrum, and covariant Markovian embeddability explicit.

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